COMP9602 — Convex Optimization

Assignment 2 (12%) [Learning Outcomes 1,2]

Due by: 23:59, Thursday, November 12 2026

Submission: please submit a softcopy through Moodle following the steps below:

1. Login Moodle.
2. Find "Assignments" in the left column and click "Assignment 2".
3. Click "Add submission", browse your solution file and save it. Done.
4. You will receive an automatic confirmation email if the submission was successful.
5. You can "Edit submission" to your already submitted file, but ONLY before the deadline.
6. Only .pdf files are accepted, and at most one file is allowed to be uploaded.

Total mark is 100.

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1. (20 marks)

Consider

$$\begin{array}{ll} \text{minimize} & \log\det X^{-1} \\ \text{subject to} & A_i^T X A_i \preceq B_i, \qquad i = 1, \ldots, m, \end{array} \tag{1}$$

with decision variable $X \in \mathbf{S}^n_{++}$, given $A_i \in \mathbb{R}^{n\times k_i}$ and $B_i \in \mathbf{S}^{k_i}_{++}$. We assume that

$$\sum_{i=1}^m A_i A_i^T \succ 0.$$

Derive the Lagrange dual of (1). Specify the dual variables and their domains, derive the dual function and its effective domain, and show that the optimal duality gap is zero.

Hint. For $X \in \mathbf{S}^n_{++}$, use

$$\log\det(X^{-1}) = -\log\det X, \qquad \nabla_X \log\det X = X^{-1}.$$

The result is given in Sec. A.4.1 (P641) of the textbook https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf.

2. (20 marks)

For $x \in \mathbb{R}^n_{++}$, define its geometric mean by

$$G(x) = \left(\prod_{i=1}^n x_i\right)^{1/n}.$$

(a) Derive the Lagrange dual function of

$$\begin{array}{ll} \text{minimize} & \dfrac{1}{n}x^T v \\ \text{subject to} & \displaystyle\sum_{i=1}^{n} \log v_i = 0, \qquad v \in \mathbb{R}^n_{++}. \end{array}$$

Note that v is the optimization variable in the above problem.

(b) Prove

$$G(x) = \inf \left\{ \frac{1}{n} v^T x : \ v \in \mathbb{R}^n_{++}, \ \prod_{i=1}^{n} v_i = 1 \right\}.$$

(c) Prove that G is concave on $\mathbb{R}^n_{++}$. Then recover the arithmetic–geometric mean inequality by choosing $v = \mathbf{1}$.

3. (20 marks)

Let $Q \in \mathbf{S}^n_{++}$, $q, a \in \mathbb{R}^n$, and $b \in \mathbb{R}$, with $a \neq 0$. Consider

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}x^T Q x + q^T x \\ \text{subject to} & a^T x \leq b. \end{array} \tag{2}$$

(a) Derive the Lagrange dual, including its dual function and effective domain.

(b) Derive explicit formulas for the optimal multiplier $\lambda^\star$ and the corresponding primal optimizer $x^\star$.

(c) For $Q = I \in \mathbf{S}^2$, $q = (-2, -1)$, $a = (1, 1)$, and $b = 1$, compute $\lambda^\star$, $x^\star$, and the common optimal value (for both primal problem and dual problem).

4. (20 marks)

Consider the convex optimization problem

$$\begin{array}{ll} \text{minimize} & f_0(x) = \displaystyle\sum_{k=1}^{n} x_k \log(x_k/y_k) \\ \text{subject to} & Ax = b \\ & \mathbf{1}^T x = 1, \end{array} \tag{3}$$

with variable $x \in \mathbb{R}^n_{++}$, parameters $y \in \mathbb{R}^n_{++}$, $A \in \mathbb{R}^{m \times n}$, and $b \in \mathbb{R}^m$. Write $A = [\, a_1 \ \cdots \ a_n \,]$ with $a_k \in \mathbb{R}^m$. Assume that feasible set of (3) contains a point in $\mathbb{R}^n_{++}$.

(a) Derive the Lagrange dual. After eliminating the multiplier for $\mathbf{1}^T x = 1$, show that the reduced dual problem is

$$\text{maximize} \quad b^T z - \log \sum_{k=1}^{n} y_k e^{a_k^T z},$$

with variable $z \in \mathbb{R}^m$. Give the full dual function and its effective domain. If $z^\star$ is optimal for the reduced dual problem, show that

$$x_k^\star = \frac{y_k e^{a_k^T z^\star}}{\sum_{j=1}^{n} y_j e^{a_j^T z^\star}}.$$

(b) Let g denote the reduced dual objective and, for $z \in \mathbb{R}^m$, define

$$p(z)_k = \frac{y_k e^{a_k^T z}}{\sum_{j=1}^n y_j e^{a_j^T z}}.$$

Show that $p(z) \succ 0$, $\mathbf{1}^T p(z) = 1$, and

$$\nabla g(z) = b - Ap(z), \qquad f_0(p(z)) - g(z) = (Ap(z) - b)^T z.$$

5. (20 marks)

Consider the convex problem

$$\begin{array}{lll} \text{minimize} & f_0(x) & \\ \text{subject to} & f_i(x) \leq 0, & i = 1, \ldots, m \\ & l_i \leq x_i \leq u_i, & i = 1, \ldots, n, \end{array} \tag{4}$$

where $f_0, \ldots, f_m$ are convex and differentiable and $l, u \in \mathbb{R}^n$ satisfy $l_i < u_i$ for all i. Let $\lambda \in \mathbb{R}^m_+$ be the multipliers for the first set of inequalities, and let $\mu, \nu \in \mathbb{R}^n_+$ be the multipliers for $x \preceq u$ and $l \preceq x$, respectively. Define

$$L(x, \lambda, \mu, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \mu^T (x - u) + \nu^T (l - x).$$

(a) Fix $x \in \mathbb{R}^n$ and $\lambda \succeq 0$. Show that one can choose $\mu, \nu \succeq 0$ so that x minimizes $L(\cdot, \lambda, \mu, \nu)$, and give explicit formulas.

(b) Let $p^\star$ denote the primal optimal value. Set $\bar{x} = (l + u)/2$ and $\lambda = 0$. Show that the resulting lower bound is

$$p^\star \geq f_0(\bar{x}) - \left(\frac{u - l}{2}\right)^T |\nabla f_0(\bar{x})|.$$

For the given numerical problem

$$\begin{array}{lll} \text{minimize} & \frac{1}{2}(x_1^2 + 4x_2^2 + x_1 x_2) - \mathbf{1}^T x & \\ \text{subject to} & \mathbf{1}^T x \leq 1, & 0 \leq x_i \leq 1, \end{array}$$

compute the dual feasible point from part (a) and the resulting lower bound of p^*.